



Advanced Time Series Analysis and Forecasting

Seminar 9: VaR, ES and backtesting: elicibility, scoring and model risk



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Acronyms used in this material (1/2)

- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **ADAPT**: Adaptive (CAViaR specification)
- ▣ **AI**: Artificial Intelligence
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **ARMA-GARCH**: ARMA mean with GARCH variance
- ▣ **AS**: Asymmetric Slope (CAViaR specification)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CEE**: Central and Eastern Europe
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- ▣ **EDF**: Empirical Distribution Function
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **ES**: Expected Shortfall
- ▣ **FHS**: Filtered Historical Simulation
- ▣ **FZ-1F**: One-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- ▣ **FZ-2F**: Two-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- ▣ **FZ0**: Zero-homogeneous Fissler–Ziegel loss (Patton, Ziegel and Chen)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ **GARCH-EDF**: GARCH with innovations from the empirical distribution function
- ▣ **GARCH-FZ**: GARCH model estimated by FZ0 minimisation
- ▣ **GARCH-N**: GARCH with Normal innovations
- ▣ **GAS-1F**: One-factor GAS model for VaR and ES
- ▣ **GAS-2F**: Two-factor GAS model for VaR and ES
- ▣ **GAS-FZ**: GAS models for VaR and ES driven by the FZ0 loss
- ▣ **GCH-FZ**: GARCH model estimated by FZ0 minimisation (table label)
- ▣ **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- ▣ **IG**: Indirect GARCH (CAViaR specification)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MCS**: Model Confidence Set
- ▣ **ML**: Machine Learning
- ▣ **PZC**: Patton, Ziegel and Chen (2019)
- ▣ **RW-125**: Rolling Window of 125 days (historical simulation)
- ▣ **RW-250**: Rolling Window of 250 days (historical simulation)
- ▣ **RW-500**: Rolling Window of 500 days (historical simulation)
- ▣ **SAV**: Symmetric Absolute Value (CAViaR specification)

Today's question and route

- **Question:** how do we score, estimate, backtest and compare forecasts of VaR and ES?
 - ▶ this seminar comes **before** Lecture 9: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: pinball consistency, ES under mixing, FZ0 by hand, durations, the square-root-of-time rule under GARCH, CAViaR recursions, ES backtests by hand, comparative zones
 - ▶ Part B: CAViaR for the S&P 500 and the BET; the replication of Patton, Ziegel and Chen; the DAX to 2026; the MCS for Bitcoin and EUR/RON; 10-day VaR of the DAX
 - ▶ Part C: conformal correction of the BET VaR; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	which forecast minimises the pinball loss; ES under mixing	Solved, Proposed	A1
A3, A4	the FZ0 loss by hand; durations between hits	Solved, Proposed	A3
A5, A6	the square-root-of-time rule under GARCH; CAViaR recursions	Solved, Proposed	A5
A7, A8	Acerbi–Székely and McNeil–Frey by hand; Nolde–Ziegel zones	Solved, Proposed	A7
B1, B2	CAViaR VaR 1% for the S&P 500 and the BET	Solved, Proposed	B1
B3, B4	Patton–Ziegel–Chen on the S&P 500; the DAX to 2026	Solved, Proposed	B3
B5, B6	MCS in stress periods; 10-day VaR of the DAX	Solved, Proposed	B5
C1, C2	conformal correction of a VaR; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
S&P 500, DAX	EODHD	daily	1990–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Bitcoin	EODHD	daily, 7 days a week	2014–2026

- Daily log returns in %
- Functions in the notebook
 - ▶ `pinball`, `fz0`, `caviar_fit`, `caviar_se`, `dq_test`, `fz_fit`, `pzc_gof`, `cp_duration_test`, `dm_test`, `mcs`, `aci_levels`
- No account or key is needed

Prerequisites for Today (1/22): VaR and ES

- On the return scale we work with the pair (v, e) : the α -quantile of returns and the mean return below it

$$v = q_\alpha, \quad \text{VaR}_\alpha = -v, \quad e = \mathbb{E}[Y \mid Y \leq v], \quad \text{ES}_\alpha = -e, \quad e \leq v < 0$$

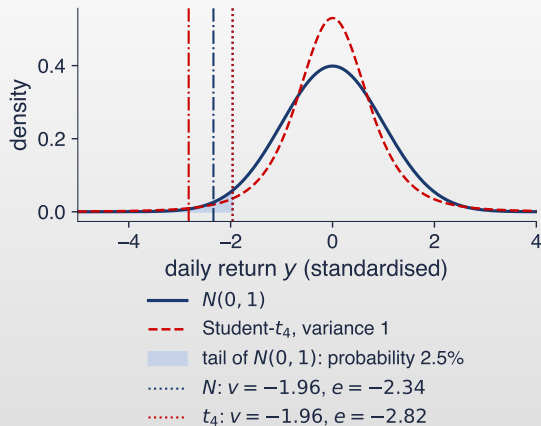
- ▶ Y : the daily return; α : the level, the target hit rate (VaR 1%: $\alpha = 0.01$); q_α : the α -quantile of Y

- Normal distribution: closed forms

$$v = \Phi^{-1}(\alpha), \quad e = -\frac{\varphi(\Phi^{-1}(\alpha))}{\alpha}$$

- ▶ Φ, φ : distribution function and density of $N(0, 1)$; for $\alpha = 2.5\%$: $(v, e) = (-1.960, -2.338)$

Prerequisites for Today (2/22): VaR and ES on a density



- Two distributions with variance 1: $N(0, 1)$ and a rescaled Student- t_4 ; $\alpha = 2.5\%$
- Shaded: the 2.5% left tail; v (dotted) is where it starts, e (dash-dot) is its centre of mass
- Same $v = -1.96$ for both, but $e = -2.34$ against -2.82
- VaR cannot see how heavy the tail is beyond it; ES can

Prerequisites for Today (3/22): VaR and ES from a volatility model

- **Location–scale model:** $y_t = \mu + \sigma_t \eta_t$, η_t i.i.d. with mean 0, variance 1

$$v_t = \mu + \sigma_t q_\alpha^\eta, \quad e_t = \mu + \sigma_t \mathbb{E}[\eta \mid \eta \leq q_\alpha^\eta]$$

- ▶ q_α^η : the α -quantile of η_t ; Normal: $\Phi^{-1}(\alpha)$; Student- t_ν with variance 1: $t_\nu^{-1}(\alpha) \sqrt{(\nu-2)/\nu}$
- ▶ EDF: q_α^η and the tail mean taken from the empirical distribution of the standardised residuals $\hat{\eta}_t = (y_t - \hat{\mu})/\hat{\sigma}_t$

- **GJR-GARCH(1,1)** [Glosten, Jagannathan and Runkle \(1993\)](#): negative shocks raise the variance more

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}\{\varepsilon_{t-1} < 0\}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- ▶ $\varepsilon_t = y_t - \mu$; $\gamma > 0$: the leverage effect

- **Filtered historical simulation (FHS)** for h days

- ▶ draw $\hat{\eta}$ with replacement from the standardised residuals, run the GARCH recursion h days ahead, sum the simulated returns
- ▶ repeat many times; the h -day VaR is minus the α -quantile of the simulated sums

Prerequisites for Today (4/22): the ten models of Patton, Ziegel and Chen

- **Rolling windows** RW-125, RW-250, RW-500: historical simulation over the last n days
 - ▶ v_t : the empirical α -quantile of $y_{t-n}, \dots, y_{t-1}$; e_t : the mean of the returns below it
- **GARCH-N, GARCH-Skt, GARCH-EDF**: ARMA-GARCH(1,1) with a location-scale VaR and ES
 - ▶ innovations: Normal, skewed Student- t [Hansen \(1994\)](#), or the empirical distribution of the residuals
- **Semiparametric models** [Patton, Ziegel and Chen \(2019\)](#), estimated by minimising the average FZ0 loss
 - ▶ GAS-2F (FZ-2F): v_t and e_t follow two recursions driven by the score of FZ0
 - ▶ GAS-1F (FZ-1F): one log-scale κ_t , $(v_t, e_t) = (a, b)e^{\kappa_t}$ (formula on the FZ0 slide)
 - ▶ GARCH-FZ: $\kappa_t^2 = 1 + \beta\kappa_{t-1}^2 + \gamma y_{t-1}^2$, $(v_t, e_t) = (a, b)\kappa_t$; Hybrid: GAS-1F plus $\delta \ln |y_{t-1}|$ in the recursion of κ_t
- Out of sample: parameters estimated once on the first window and kept fixed

Prerequisites for Today (5/22): scores and elicibility

- A score $S(x, y)$ (the loss of forecast x when y is observed) is **consistent** for the functional T if the truth minimises the expected score [Gneiting \(2011\)](#)

$$\mathbb{E}_F S(T(F), Y) \leq \mathbb{E}_F S(x, Y) \quad \text{for all } x$$

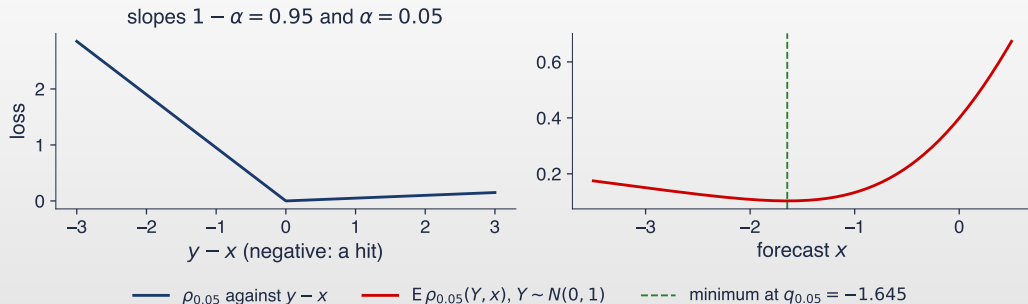
- ▶ $T(F)$: a number computed from the distribution F (quantile, ES, mean); T is **elicitable** if a strictly consistent S exists
- ▶ necessary condition: convex level sets (a mixture of two distributions with the same functional value keeps that value)

- Pinball loss, consistent for the quantile q_α

$$\rho_\alpha(y, x) = (1\{y \leq x\} - \alpha)(x - y) \geq 0$$

- ▶ $1\{\cdot\}$: indicator (1 if true, 0 otherwise); on a hit day ($y \leq x$) the loss is $(1 - \alpha)(x - y)$, otherwise $\alpha(y - x)$

Prerequisites for Today (6/22): the pinball loss and its expectation



- Left: $\rho_{0.05}$ is a tilted absolute value; a hit (negative $y - x$) costs 19 times more per unit than a miss
- Right: for $Y \sim N(0, 1)$ the expected loss is minimised exactly at the 5% quantile -1.645 : the loss is consistent for the quantile

Prerequisites for Today (7/22): the FZ0 loss

- ES alone is not elicitable; the pair (VaR, ES) is [Fissler and Ziegel \(2016\)](#). **FZ0** [Patton, Ziegel and Chen \(2019\)](#) is the scale-free loss for the pair

$$L(y, v, e) = -\frac{1\{y \leq v\}(v - y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$

- ▶ first term: positive only on hit days, proportional to how far y falls below v (note $e < 0$); v/e and $\ln(-e)$: penalise a too small or too large ES
- ▶ lower average FZ0 = better forecast of the pair; values can be negative

- GAS-1F** [Patton, Ziegel and Chen \(2019\)](#): one latent log-scale κ_t moves both risk measures; estimated by FZ0 minimisation

$$(v_t, e_t) = (a, b) e^{\kappa_t}, \quad \kappa_t = \beta \kappa_{t-1} + \gamma \frac{-1}{e_{t-1}} \left(\frac{1\{y_{t-1} \leq v_{t-1}\} y_{t-1}}{\alpha} - e_{t-1} \right)$$

- ▶ $b < a < 0$: VaR and ES at $\kappa = 0$; β : persistence; γ : reaction, active mainly on hit days

Prerequisites for Today (8/22): the expected FZ0 loss



- ▣ $Y \sim N(0, 1)$, $\alpha = 2.5\%$; true pair $(v, e) = (-1.960, -2.338)$
- ▣ Moving v (left) or e (right) away from the truth always raises the expected FZ0 loss
- ▣ The curves are asymmetric: a too small $|e|$ is punished more than a too large one

Prerequisites for Today (9/22): CAViaR and the DQ test

- **CAViaR** Engle and Manganelli (2004): autoregressive models for $\text{VaR}_t > 0$, with y_{t-1} yesterday's return

$$\text{SAV: } \beta_1 + \beta_2 \text{VaR}_{t-1} + \beta_3 |y_{t-1}|, \quad \text{AS: } \beta_1 + \beta_2 \text{VaR}_{t-1} + \beta_3 y_{t-1}^+ + \beta_4 y_{t-1}^-$$

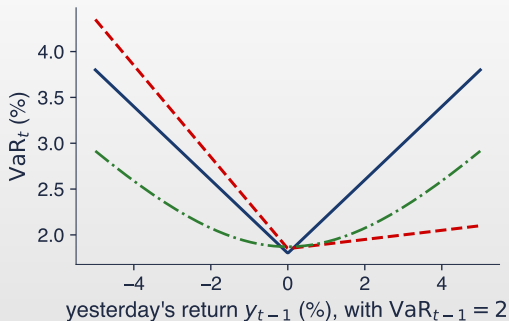
- ▶ IG: $(\beta_1 + \beta_2 \text{VaR}_{t-1}^2 + \beta_3 y_{t-1}^2)^{1/2}$; adaptive: VaR is raised after a hit; $y^+ = \max(y, 0)$, $y^- = \max(-y, 0)$
- ▶ β_2 : persistence; β_3, β_4 : reactions to gains and losses; estimation: minimise the average pinball loss of $q_t = -\text{VaR}_t$ from many random starts

- **DQ test**: regress the centred hits on information known at $t - 1$ and test that all coefficients are zero

$$\text{Hit}_t = 1\{y_t < q_t\} - \alpha, \quad \text{DQ} = \frac{\text{Hit}' X (X' X)^{-1} X' \text{Hit}}{\alpha(1 - \alpha)} \sim \chi_6^2$$

- ▶ X : matrix whose rows are $(1, \text{the four lags of } \text{Hit}_t, q_t)$; the χ_6^2 reference holds out of sample; a large DQ rejects calibration

Prerequisites for Today (10/22): CAViaR news-impact curves



- SAV: $0.20 + 0.80 \text{VaR}_{t-1} + 0.40 |y_{t-1}|$
- - AS: $0.15 + 0.85 \text{VaR}_{t-1} + 0.05 y^+ + 0.50 y^-$
- · - IG: $(0.10 + 0.85 \text{VaR}_{t-1}^2 + 0.20 y^2)^{1/2}$

- VaR_t as a function of yesterday's return, with $\text{VaR}_{t-1} = 2$; illustrative coefficients
- SAV and IG: symmetric, gains and losses of equal size raise VaR equally
- AS: a loss raises VaR ten times more than a gain ($\beta_4/\beta_3 = 10$)
- Adaptive CAViaR:

$$\text{VaR}_t = \text{VaR}_{t-1} + \beta_1 [1\{y_{t-1} < -\text{VaR}_{t-1}\} - \alpha]$$
 - the indicator is smoothed by a logistic function in estimation

Prerequisites for Today (11/22): VaR backtests

- Counting tests [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
 - ▶ Kupiec: likelihood-ratio (LR) test of the hit rate against α , χ^2_1 ; Christoffersen: adds dependence between consecutive hits, χ^2_2
- **Durations** [Christoffersen and Pelletier \(2004\)](#): under a correct VaR the gaps between hits are geometric with mean $1/\alpha$
 - ▶ Weibull alternative with shape b : $b = 1$ no memory, $b < 1$ clustering (a hit is followed by another soon)
- PZC calibration regressions: the two standardised generalised residuals

$$1\{y_t \leq v_t\} - \alpha, \quad \frac{1\{y_t \leq v_t\} y_t}{\alpha e_t} - 1$$

- ▶ each regressed on $(1, \text{its own first lag, } v_t \text{ or } e_t)$; Wald test of three zero coefficients, χ^2_3

Prerequisites for Today (12/22): counting tests: the formulas

- **Kupiec Kupiec (1995)**: unconditional coverage; N hits in T days, $\hat{\pi} = N/T$

$$LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-N} \alpha^N}{(1 - \hat{\pi})^{T-N} \hat{\pi}^N} \sim \chi_1^2$$

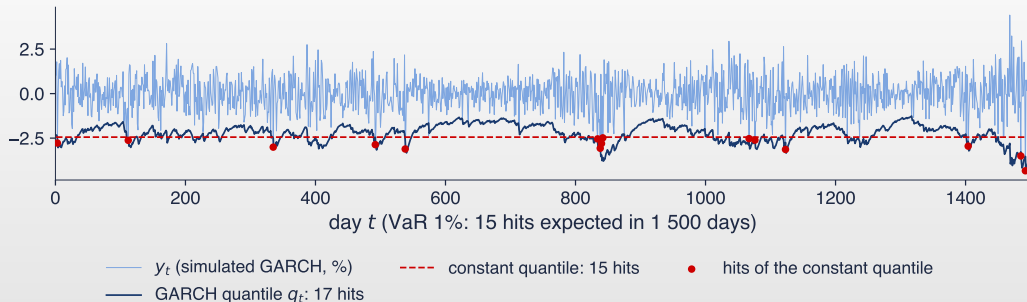
- ▶ 0 when $\hat{\pi} = \alpha$; grows with too few and with too many hits

- **Christoffersen Christoffersen (1998)**: independence; n_{ij} : number of days with state i followed by state j ($1 = \text{hit}$)

$$LR_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}}, \quad \hat{\pi}_{i1} = \frac{n_{i1}}{n_{i0} + n_{i1}}$$

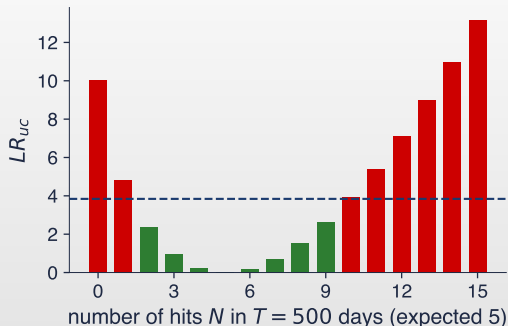
- ▶ $\hat{\pi}_{11} > \hat{\pi}_{01}$: a hit makes another hit tomorrow more likely (clustering)
- ▶ conditional coverage: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi_2^2$

Prerequisites for Today (13/22): hits of a dynamic and of a constant VaR



- Simulated GARCH returns, VaR 1%; blue: the true GARCH quantile q_t ; red: one constant quantile with exactly 15 hits
- The constant quantile has the right number of hits, but they cluster in turbulent spells: Kupiec passes, independence and DQ tests can reject

Prerequisites for Today (14/22): the Kupiec acceptance region



--- χ^2_1 critical value 3.84

■ not rejected at 5%

■ rejected at 5%

- $T = 500$ days, VaR 1%: 5 hits expected
- Green: numbers of hits not rejected at 5%, from 2 to 9
- The region is wide: with few expected hits the test has little power
- Zero hits is also rejected: a too conservative VaR fails too

Prerequisites for Today (15/22): durations between hits: the formulas

- **Under a correct VaR** Christoffersen and Pelletier (2004): hits are i.i.d. Bernoulli(α), so the gap D between two hits is geometric

$$\Pr(D = d) = \alpha(1 - \alpha)^{d-1}, \quad \mathbb{E}D = \frac{1}{\alpha}, \quad \Pr(D \leq k) = 1 - (1 - \alpha)^k$$

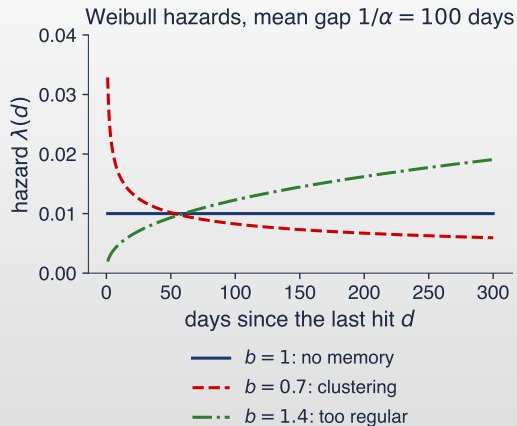
- ▶ continuous approximation: exponential with constant hazard λ ; its ML estimate is $\hat{\lambda} = 1/\bar{D}$, $\bar{D}$: the mean gap

- **Weibull alternative**: the hazard depends on the time since the last hit

$$f(d) = b\lambda^b d^{b-1} e^{-(\lambda d)^b}, \quad \lambda(d) = \frac{f(d)}{1 - F(d)} = b\lambda^b d^{b-1}$$

- ▶ $\lambda(d)$: the probability rate of a hit after d quiet days; b : the shape, λ : the scale
- ▶ test $H_0: b = 1$ (exponential) by an LR test, χ^2_1

Prerequisites for Today (16/22): Weibull hazards



- Three Weibull hazards with the same mean gap of 100 days (VaR 1%)
- $b = 1$: constant hazard α , the correct VaR
- $b < 1$: right after a hit another one is likely; long quiet periods follow
- $b > 1$: hits come too regularly, rarely seen in practice

Prerequisites for Today (17/22): ES backtests

- In loss units: $L_t = -y_t$, VaR_t , $\text{ES}_t > 0$, hit indicator $I_t = 1\{L_t > \text{VaR}_t\}$
- Acerbi and Székely (2014): the Z_2 statistic compares realised tail losses with the forecast ES

$$Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$$

- ▶ T : number of days; $\mathbb{E}Z_2 = 0$ under H_0 ; negative if risk is underestimated (too many or too large losses)
- McNeil and Frey (2000): standardised exceedance residuals on hit days

$$r_t = \frac{L_t - \text{ES}_t}{\sigma_t}$$

- ▶ σ_t : forecast volatility; mean 0 under H_0 ; a positive mean means ES too small

Prerequisites for Today (18/22): ES backtests: Z_1 and the t statistic

- Z_1 Acerbi and Székely (2014): the size of the losses given a hit; $N = \sum_t l_t$ hit days

$$Z_1 = 1 - \frac{1}{N} \sum_t \frac{L_t l_t}{\text{ES}_t}$$

- ▶ $\mathbb{E}Z_1 = 0$ under H_0 ; negative if hit-day losses exceed the forecast ES; unlike Z_2 it ignores how many hits there are

- McNeil–Frey t statistic McNeil and Frey (2000): the mean exceedance residual over its standard error

$$t = \frac{\bar{r}}{s_r / \sqrt{N}}, \quad \bar{r} = \frac{1}{N} \sum_{t: l_t=1} r_t, \quad s_r^2 = \frac{1}{N-1} \sum_{t: l_t=1} (r_t - \bar{r})^2$$

- ▶ one-sided: a large positive t means ES too small; with few hits use the bootstrap rather than $N(0, 1)$
- Critical values of Z_1 and Z_2 are simulated under H_0 (they depend on T , α and the tail)

Prerequisites for Today (19/22): comparison and horizon

- DM Diebold and Mariano (1995): t statistic of the mean loss difference, with a Newey–West variance (Chapter 1)
 - ▶ Nolde–Ziegel zones Nolde and Ziegel (2017): one-sided tests at 5% of internal minus standard; green if significantly better, red if significantly worse, yellow otherwise
- MCS Hansen, Lunde and Nason (2011): eliminate the worst model until equal predictive ability is not rejected
 - ▶ MCS p-value ≥ 0.10 : the model is in the 90% set
- Horizon: $\sqrt{h}$ scaling is exact only for i.i.d. Normal returns; under GARCH(1,1) the expected h -day variance is

$$\mathbb{E}_t \sum_{k=1}^h \sigma_{t+k}^2 = \sum_{k=0}^{h-1} [\bar{\sigma}^2 + (\alpha + \beta)^k (\sigma_{t+1}^2 - \bar{\sigma}^2)], \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

- ▶ ω, α, β : GARCH parameters (here α is the ARCH coefficient, not the VaR level); $\bar{\sigma}^2$: unconditional variance
- ACI Gibbs and Candès (2021): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\text{err}_t = 1$ if the model's quantile at level α_t is hit; γ : step; long-run hit frequency $\rightarrow \alpha$ for any data

Prerequisites for Today (20/22): Diebold–Mariano and the MCS: the formulas

- **DM Diebold and Mariano (1995):** loss differential $d_t = L_t^A - L_t^B$ of models A and B (e.g. FZ0 losses)

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}/T}} \approx N(0, 1), \quad \hat{V} = \hat{\gamma}_0 + 2 \sum_{j=1}^m \left(1 - \frac{j}{m+1}\right) \hat{\gamma}_j$$

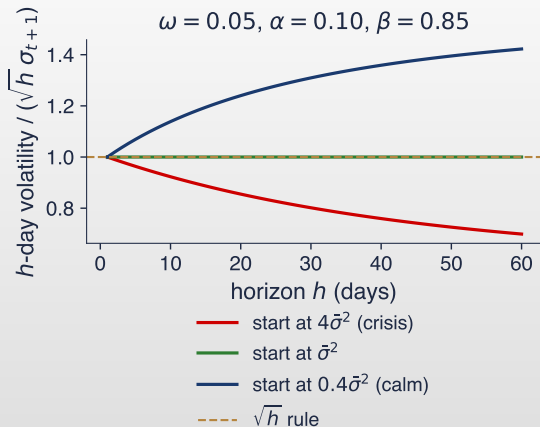
- ▶ $\bar{d}$: the mean of d_t ; $\hat{\gamma}_j$: the sample autocovariance of d_t at lag j ; m : the number of lags (Newey–West)
- ▶ $DM < 0$: A has the lower average loss; $|DM|$ large relative to $N(0, 1)$: the difference is significant

- **MCS Hansen, Lunde and Nason (2011):** H_0 = equal expected loss of all models in the current set $\mathcal{M}$

$$T_{\max} = \max_{i \in \mathcal{M}} t_i, \quad t_i = \frac{\bar{d}_{i\cdot}}{\widehat{\text{se}}(\bar{d}_{i\cdot})}, \quad \bar{d}_{i\cdot} = \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \bar{d}_{ij}$$

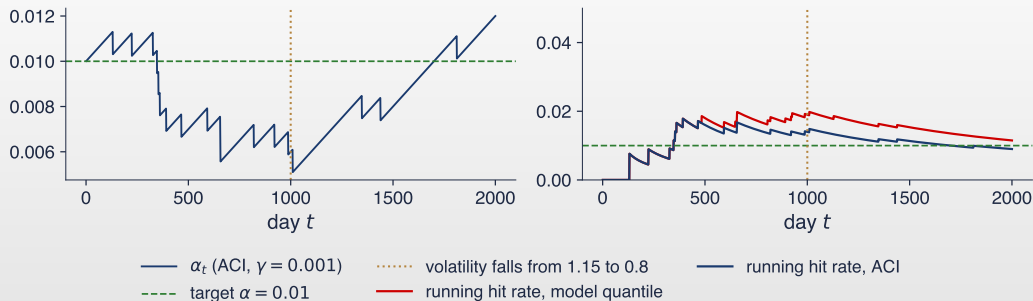
- ▶ $\bar{d}_{ij}$: mean loss of i minus that of j ; the distribution of $T_{\max}$ comes from a block bootstrap
- ▶ if rejected, remove the model with the largest t_i and repeat; the survivors form the MCS

Prerequisites for Today (21/22): the square-root-of-time rule under GARCH



- Ratio of the true h -day volatility to $\sqrt{h} \sigma_{t+1}$, GARCH(1,1), Normal shocks
- From a crisis (σ_{t+1}^2 high) the variance reverts down: $\sqrt{h}$ overstates
- From a calm day it reverts up: $\sqrt{h}$ understates
- Only at $\sigma_{t+1}^2 = \bar{\sigma}^2$ is the rule exact for the variance (fat tails and leverage still change the quantile)

Prerequisites for Today (22/22): adaptive conformal inference



- Simulated: the model always uses $N(0, 1)$, the true volatility is 1.15, then 0.8; VaR 1%, ACI with $\gamma = 0.001$
- Left: α_t falls while the model has too many hits and rises after the volatility falls
- Right: the running hit rate of ACI returns to 1%; that of the fixed model quantile does not

A1: the pinball loss and its minimiser [Solved]

Task

- ▣ Ten returns:
 $-3.1, -1.2, -0.4, 0.3, 0.9, 1.5, 2.2, -2.0, 0.1, 0.6$; $\alpha = 0.1$.
- ▣ 1. Show that $\frac{d}{dx} \mathbb{E} \rho_\alpha(Y, x) = F(x) - \alpha$.
- ▣ 2. Compute the average pinball loss at
 $x = -3.1, -2.6, -2.0, -1.2$.
- ▣ 3. Which forecasts minimise it, and why is the minimiser not unique?
- ▣ Report: one derivation; four numbers; an interval and one sentence.

Solution

- ▣ 1. $\mathbb{E} \rho = (1 - \alpha) \int_{-\infty}^x (x - y) dF + \alpha \int_x^{\infty} (y - x) dF$; derivative
 $(1 - \alpha)F(x) - \alpha(1 - F(x)) = F(x) - \alpha$
- ▣ 2. 0.299, 0.299, 0.299, 0.379
- ▣ 3. Every $x \in [-3.1, -2.0]$: there $F_n(x) = 0.1 = \alpha$ and the empirical objective is flat
- ▣ Interpretation: with $n\alpha$ an integer, the sample quantile is an interval; with few tail points the loss barely separates forecasts

A2: ES is not elicitable [Proposed]

Solution discussed in class

Task

- ▣ $F_0 = N(0, 1)$ and $F_1 = t(3)$ scaled by s so that both have ES 2.5% equal to that of $N(0, 1)$. Model: A1.
- ▣ 1. Using $e_t = -s t_3(q)(3 + q^2)/(2\alpha)$ at $q = t_3^{-1}(\alpha)$, find s .
- ▣ 2. For the mixture $0.4F_0 + 0.6F_1$, compute its 2.5% quantile and its tail mean (notebook section A2).
- ▣ 3. Explain why this proves that no strictly consistent loss exists for ES alone.
- ▣ Report: one number; two numbers; three sentences.

A3: the FZ0 loss by hand [Solved]

Task

- ▣ $\alpha = 0.025$; forecast 1: $(v, e) = (-1.96, -2.34)$; forecast 2: $(-2.5, -3.2)$.
- ▣ 1. Compute FZ0 for both forecasts at $y = -3$ and at $y = 0.5$.
- ▣ 2. Which forecast is better on a hit day, and which on a calm day?
- ▣ 3. If $Y \sim N(0, 1)$, which forecast has the lower expected loss (notebook: 10^6 draws)?
- ▣ Report: four numbers; two sentences; two numbers.

Solution

- ▣ 1. $y = -3$: 18.466 and 7.194; $y = 0.5$: 0.688 and 0.944
- ▣ e.g. forecast 1 at $y = -3$: $-\frac{1.04}{0.025 \cdot (-2.34)} + \frac{-1.96}{-2.34} + \ln 2.34 - 1$
- ▣ 2. The conservative forecast 2 wins on the hit day, forecast 1 on the calm day: a single day says nothing
- ▣ 3. $\mathbb{E}L_1 = 0.8486$, $\mathbb{E}L_2 = 0.9689$: forecast 1 is the true $(v, e) = (-1.960, -2.338)$ and wins on average

A4: durations between hits [Proposed]

Solution discussed in class

Task

- ▣ VaR 1%; twelve durations (days) between consecutive hits: 3, 41, 2, 160, 95, 1, 4, 230, 12, 2, 68, 7. Model: A3.
- ▣ 1. Compute the mean duration and the maximum-likelihood hazard of the exponential model; compare with α .
- ▣ 2. Compute the share of durations of at most 5 days and its value under a correct VaR, $1 - 0.99^5$.
- ▣ 3. Would a Weibull fit give $b < 1$ or $b > 1$? Explain.
- ▣ Report: two numbers and one sentence; two numbers; one sentence.

A5: the square-root-of-time rule under GARCH [Solved]

Task

- ▣ GARCH(1,1): $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$; Normal innovations; horizon $h = 10$.
- ▣ 1. Compute the unconditional variance $\bar{\sigma}^2$.
- ▣ 2. For $\sigma_{t+1}^2 = 4$ (crisis) and 0.5 (calm), compute $\sum_{k=0}^9 \mathbb{E}_t \sigma_{t+1+k}^2$ and the ratio $\sqrt{\text{sum}/(10\sigma_{t+1}^2)}$.
- ▣ 3. When does $\sqrt{10}$ scaling overstate the 10-day volatility?
- ▣ Report: one number; four numbers; one sentence.

Solution

- ▣ 1. $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.00$
- ▣ 2. Crisis: sum 37.44, ratio 0.967; calm: sum 5.43, ratio 1.042
- ▣ 3. When today's variance exceeds $\bar{\sigma}^2$: volatility is expected to fall; with persistence 0.98 the effect over 10 days is a few percent
- ▣ Interpretation: mean reversion alone moves the 10-day VaR by a few percent; fat tails and leverage move it more

A6: CAViaR recursions [Proposed]

Solution discussed in class

Task

- ▣ SAV $(\beta_1, \beta_2, \beta_3) = (0.36, 0.70, 0.60)$; AS $(0.26, 0.81, 0.04, 0.62)$; $\text{VaR}_{t-1} = 2.0$; $y_{t-1} = -3.0$, then $y_t = 1.0$. Model: A5.
- ▣ 1. Compute VaR_t and VaR_{t+1} for SAV and for AS.
- ▣ 2. Compute the AS value of VaR_t after $y_{t-1} = +3.0$ instead.
- ▣ 3. Interpret β_4/β_3 and β_2 .
- ▣ Report: four numbers; one number; two sentences.

A7: ES backtests by hand [Solved]

Task

- $T = 250$, ES 2.5%; seven hit days with loss L , VaR, ES, σ (%): (3.9, 2.6, 3.3, 1.2), (2.8, 2.5, 3.1, 1.1), (4.6, 2.9, 3.6, 1.3), (3.1, 2.7, 3.4, 1.2), (6.2, 3.0, 3.8, 1.4), (2.9, 2.6, 3.2, 1.15), (3.4, 2.8, 3.5, 1.25).
- 1. Compute Z_2 and $Z_1 = 1 - \frac{1}{N} \sum L_t l_t / \text{ES}_t$.
- 2. Compute the McNeil–Frey residuals, their mean and the t statistic.
- 3. Compare with the reference 5% critical value of Z_2 , about -0.70 for $T = 250$.
- Report: two numbers; seven residuals and two numbers; one sentence.

Solution

- 1. $\sum L/\text{ES} = 7.784$; $Z_2 = 1 - 7.784/6.25 = -0.245$; $Z_1 = -0.112$
- 2. Residuals 0.500; -0.273 ; 0.769 ; -0.250 ; 1.714 ; -0.261 ; -0.080 ; mean 0.303 , $t = 1.07$
- 3. Z_2 is negative but above -0.70 : seven hits instead of 6.25 and one large loss are not enough to reject
- Interpretation: Z_2 mixes frequency and size; Z_1 and McNeil–Frey look only at size given a hit, with seven observations

A8: comparative backtest zones [Proposed]

Solution discussed in class

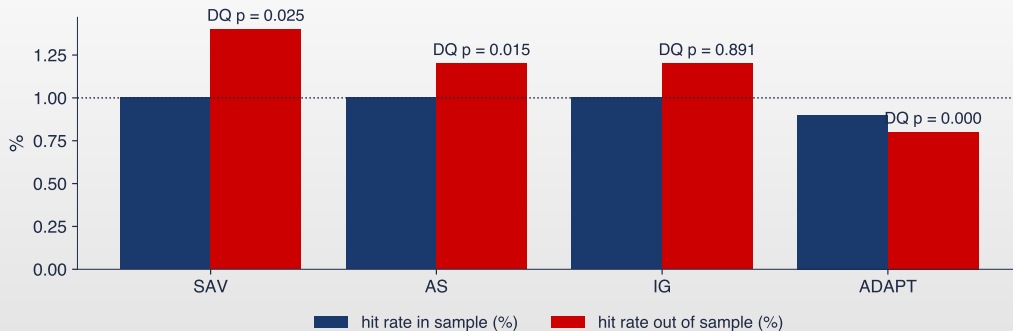
Task

- ▣ Three internal models A, B, C against a standard model; DM statistics of internal minus standard average FZ0 loss: -2.31 , 0.84 , 2.05 . Model: A7.
- ▣ 1. Give the one-sided 5% critical value.
- ▣ 2. Assign each model to the green, yellow or red zone.
- ▣ 3. Explain why a yellow model is not “as good as” the standard.
- ▣ Report: one number; three zones; one sentence.

B1: CAViaR for the S&P 500 [Solved]

- **Question:** which CAViaR specification gives a calibrated VaR 1% for the S&P 500 out of sample?
- **Inputs:** the last 3,392 daily returns to 18 September 2026: 2,892 for estimation, 500 out of sample (the design of Engle and Manganelli 2004)
- **Tasks**
 1. Estimate SAV, AS, IG and the adaptive model with the Engle–Manganelli search.
 2. Report the coefficients with their asymptotic standard errors.
 3. Report the out-of-sample hit rates and the out-of-sample DQ test.
 4. Interpretation: which model would you use, and how much would you trust the test?
- **Report:** a table of coefficients, four hit rates, four DQ statistics and p -values, one sentence
- **Notebook:** section B1

B1: solution (1/2) [Solved]



B1: solution (2/2) [Solved]

Model	β_1	β_2	β_3	β_4	DQ (p)
SAV	0.359 (0.217)	0.702 (0.118)	0.604 (0.243)	–	14.43 (0.025)
AS	0.264 (0.137)	0.805 (0.080)	0.044 (0.185)	0.616 (0.149)	15.81 (0.015)
IG	0.942 (0.467)	0.674 (0.105)	1.429 (0.702)	–	2.29 (0.891)
ADAPT	1.190 (0.069)	–	–	–	27.50 (<0.001)

- Out-of-sample hit rates: SAV 1.4%, AS 1.2%, IG 1.2%, adaptive 0.8% (5 expected hits in 500 days)
- Interpretation: IG passes DQ, SAV and AS fail at 5%, the adaptive model clearly; with about five hits the decision rests on the April 2025 cluster: use IG or AS, and report the test as fragile

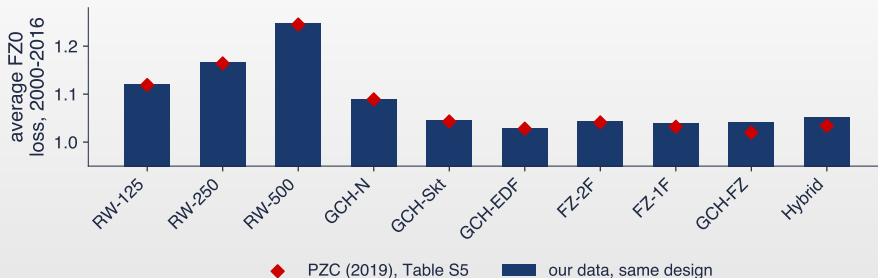
B2: CAViaR for the BET [Proposed]

- **Question:** does the asymmetric slope also describe the tail of the Bucharest market?
- **Inputs:** BET, the last 3,392 daily returns (21 February 2013 – 18 September 2026), out of sample from 12 September 2024; model: B1
- **Tasks**
 1. Estimate SAV and AS for VaR 1% and report the coefficients with standard errors.
 2. Estimate a GARCH(1,1) with Student-t innovations on the same window and compute its VaR 1%.
 3. Compare the three out of sample: hit rate, DQ p -value and average pinball loss.
 4. Interpretation: is the BET tail asymmetric, and does the asymmetry pay off out of sample?
- **Report:** two coefficient vectors with standard errors, a table of nine numbers, one sentence
- **Notebook:** section B2

B3: replicating Patton, Ziegel and Chen [Solved]

- **Question:** do we reproduce the out-of-sample ranking of their Table S5 ($\alpha = 2.5\%$) on our data?
- **Inputs:** S&P 500, estimation January 1990 – December 1999, parameters fixed for 2000–2016 ($T = 4\,275$); the ten models of the paper
- **Tasks**
 1. Estimate the rolling-window, ARMA-GARCH and FZ0 models as in the paper.
 2. Compute the average out-of-sample FZ0 loss of each model and compare it with the paper.
 3. Compute DM statistics of GARCH-EDF, RW-250 and GARCH-N against GAS-1F.
 4. Interpretation: which conclusions of the paper survive on our data?
- **Report:** ten pairs of losses, three DM statistics, one sentence
- **Notebook:** section B3

B3: solution [Solved]



- Ours (paper): RW-125 1.120 (1.119), RW-500 1.247 (1.245), GARCH-N 1.089 (1.089), GARCH-EDF 1.028 (1.028), GAS-1F 1.038 (1.032), GARCH-FZ 1.040 (1.020)
- DM against GAS-1F: GARCH-EDF 0.52, RW-250 3.87, GARCH-N 1.95
- Interpretation: the losses match to about two hundredths; rolling windows are clearly worse, while GARCH-EDF, GAS-1F and GARCH-FZ cannot be separated: the paper's main message survives, the identity of the winner does not

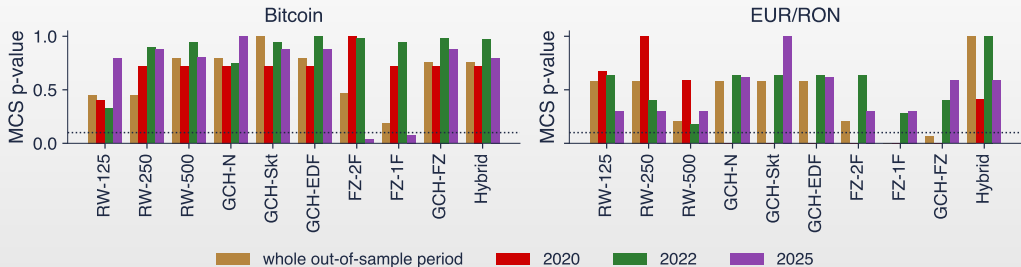
B4: the DAX to 2026 [Proposed]

- **Question:** do the semiparametric models keep their advantage on another index and a longer period?
- **Inputs:** DAX, estimation 1990–1999, out of sample January 2000 – September 2026 ($T = 6\,787$), $\alpha = 2.5\%$; model: B3
- **Tasks**
 1. Report the average FZ0 loss and the hit rate of RW-250, GARCH-Skt, GARCH-EDF and GAS-1F.
 2. Run the PZC goodness-of-fit regressions for VaR and ES.
 3. Compute the DM statistic of GAS-1F against GARCH-EDF on the whole period, before 2017 and from 2017.
 4. Interpretation: what does the DAX add to the S&P 500 evidence?
- **Report:** a table of four models with losses, hit rates and two p -values, three DM statistics, one sentence
- **Notebook:** section B4

B5: the MCS in stress periods [Solved]

- **Question:** can the crisis windows separate ES models for Bitcoin and EUR/RON?
- **Inputs:** the ten PZC models, $\alpha = 2.5\%$; Bitcoin out of sample from 1 January 2020 ($T = 2\,453$), EUR/RON from 1 July 2015 ($T = 2\,815$); stress periods 2020, 2022, 2025
- **Tasks**
 1. Compute the 90% MCS of the FZ0 losses on the whole out-of-sample period (blocks of 10 days, 1000 replications).
 2. Repeat the computation in each stress period (blocks of 5 days).
 3. Count the GAS-1F hits in each period.
 4. Interpretation: what does the size of the MCS say?
- **Report:** MCS sizes and members, numbers of hits, one sentence
- **Notebook:** section B5

B5: solution [Solved]



- Whole period: Bitcoin 10 of 10 models in the MCS (eliminated: none); EUR/RON 8 (eliminated: FZ-1F, GCH-FZ)
- Bitcoin, MCS size (days, hits): 2020 10 (133, 3), 2022 10 (362, 10), 2025 8 (120, 1)
- EUR/RON in 2020: 4 models survive (RW-125, RW-250, RW-500, Hybrid) with 0 GAS-1F hits; 2022 and 2025: all ten
- Interpretation: a few tail days cannot rank ten models; the 2020 EUR/RON result comes from over-predicted risk, not from hits

B6: 10-day VaR of the DAX [Proposed]

- **Question:** how wrong is the square-root-of-time rule for the DAX, and can a backtest tell?
- **Inputs:** DAX 2001–2026; GJR-GARCH(1,1), rolling 2000-day window, re-estimated every 250 days; 2000 bootstrapped 10-day paths; model: B5
- **Tasks**
 1. Compute every day the 10-day VaR 1% by FHS and $\sqrt{10}$ times the 1-day VaR 1%.
 2. Report the median ratio, and its median on the 20% calmest and 20% most volatile days.
 3. Backtest both on non-overlapping 10-day windows (hits, Kupiec p).
 4. Interpretation: is the difference economically or statistically detectable?
- **Report:** three ratios, two hit counts and two p -values, one sentence
- **Notebook:** section B6

C1: a conformal correction for the BET VaR [Proposed]

- ▣ **Question:** can adaptive conformal inference repair a miscalibrated Normal GARCH VaR 1% for the BET, and at what cost?
- ▣ **Inputs:** BET 2016–2026 ($T = 2\,675$ days); GARCH-N, rolling 1000-day window, re-estimated every 250 days; model: B1
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypothesis, models, step sizes γ , sample, what would change your mind.
 2. Apply ACI with $\gamma = 0.001, 0.005, 0.02$ and report the hit rates, Kupiec and conditional-coverage p -values, and the range of α_t .
 3. Interpretation: what does ACI fix, and what does it not fix?
- ▣ **Report:** a five-line plan, a table of three rows, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise risk forecasting. The answer:
- (a) VaR 99% is the 99th percentile of returns; for a standard Normal it equals 2.33.
- (b) ES can be compared across models with the mean squared error of the ES forecasts.
- (c) The pinball loss is strictly consistent for the alpha-quantile.
- (d) For a standard Normal, ES 2.5% equals 1.96, the same as the 2.5% quantile.
- (e) Estimating the model on more data before a backtest can only make the test more reliable, so the fixed window does not matter.
- (f) The 10-day VaR is always $\sqrt{10}$ = 3.16 times the 1-day VaR.
- (g) If several models are in the MCS, they are equally good.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Score a quantile with the pinball loss and the pair (VaR, ES) with FZ0; never score ES alone
- ▣ CAViaR estimates the tail directly; asymmetry matters for equity indices
- ▣ Published average losses replicate; published tests are fragile
- ▣ Few tail days: wide MCS, low power, estimation risk; report uncertainty
- ▣ An AI answer is a draft: check conventions, consistency of the loss and the horizon

After the seminar

- Lecture 9 develops each topic of today: elicibility, CAViaR and GAS-FZ models, backtesting with estimation risk, comparison, horizon and model risk
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: calibrated tail forecasts for Central and Eastern European markets
- Reading: [Engle and Manganelli \(2004\)](#); [Patton, Ziegel and Chen \(2019\)](#); [Fissler and Ziegel \(2016\)](#); [Nolde and Ziegel \(2017\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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